LOCATION PLANNING

Reference Text:

- William Stevenson, Operations Management —13th ed., McGraw Hill Education, NY
  - *Chapter-8: Location Planning and Analysis*

Why Location Matters

- Location of a Property (Land, Building, Farm etc.)
- Location of an ATMs
- Location of a supermarket
- Location of food outlets
- Location of a steel plant
- Location of Hydroelectric Power Plant
- Location of water treatment plants
- Location of a retail store
- Location of a school, college, hospital etc.

Physical locations should be visible and accessible as per demand density

Web-based businesses less dependent on location decisions; they can exist just about anywhere.

Levels of Location Decisions

Location Decisions: Site Selection

- Cost of the land/land development
- Room for future expansion
- Power and water availability
- Drainage and Waste management
- Transportation Infrastructure: Access Roads, Rail spurs, Air freight, Sea routes etc.
- Environmental zoning restrictions etc.

Location Decisions: Region Selection

1. **Proximity to Raw Materials (Why so ?)**
   - ***Necessity:*** *Mining, Farming, Forestry etc.*
   - ***Perishability*** *of raw materials: Food Processing Firms*
   - *Cost of Transportation*
2. **Proximity to Markets (Why so?)**
   - ***Customer accessibility/convenience****: Service firms, banks, hospitals, schools, government offices etc.*
   - ***Perishability*** *of the finished product: bakeries, flower shops, and fresh seafood stores etc.*
   - *Cost of distribution*
3. **Availability of skilled & motivated workforce**

Location Decisions: Identifying a Country

1. **Buyer's sentiments towards national brands (Made in India vs Made in USA ?)**
2. **Government Regulations & Incentives**
3. **Skilled & Motivated Workforce**
4. **Technology (IT & Others) Infrastructure**
5. **Market Growth Potential**
6. **Safety**

Evaluating Location Alternatives: **Qualitative Approaches**

- **Industry Precedence**
  - ***Since it has worked for the past similar industry, it will work for my company as well***
  - ***Limitations: Emerging opportunities can't be considered***
- **Supply Chain Factor**
  - ***In manufacturing sector, locations are primarily dependent on supply-side locations***
  - ***In service sector, locations are primarily dependent on demand-side (market area)***
- **Preferential Factor**
  - ***My company shoukd be located in my hometown only so that I can create employment here.***
  - ***Other factors (e.g., investor's preference, government's future policies e.g., 'make in India', etc)***

Evaluating Location Alternatives:

- Quantitative Approaches
  - *Factor Rating*
  - *The Centre of Gravity Method*
  - *Cost-Volume-Profit Analysis (Break-Even Analysis)*
  - *Others classical approaches such as transportation model, simulations, etc.*
- **Advanced Methods; GIS/AI -based Location heat maps based on** footfalls, demand density, etc. for
  - Locating Dark Stores, Hyperlocal Warehouses (3-5 km delivery radiuses),
  - Locating in Commercial Spaces
  - Locating in FSSAI complaint spaces (Schedule-4 regulations) for food business.

Factor Rating Method

Identify the relevant factors

e.g.,
F1: market proximity,
F2: labour availability,
F3: operating cost, etc,

Assign weights to those individual factors

e.g., (w1, w1, w3, etc. assigned to F1, F2, F3 respectively)

Score each factor on a common scale (1-100)

Decide a minimum acceptable limit for each factor (e.g., 50 out of 100)

Calculate the composite (weighted factor) **score each location alternative**

i.e.,
Composite Score for location L1 = …
(w1*F1 +w2*F2+…)
Composite Score for location L2 = …

Decide a minimum acceptable limit for each factor

i.e.,
F >= 80/100

Select the alternative having highest composite score

e.g.,
Score of L1 = 87/100
Score of L2 =92/100

Factor Rating Method of Evaluating Location Alternatives

A photo-processing company intends to open a new branch store. The following table contains information on two potential locations. Which is the better alternative?

		Scores (Out of 100)		Weighted Scores	
Factor	**Weight**	**Alt. 1**	**Alt. 2**	**Alternative 1**	**Alternative 2**
Proximity to existing store	.10	100	60		
Traffic volume	.05	80	80		
Rental costs	.40	70	90	.40(70) = 28.0	.40(90) = 36.0
Size	.10	86	92	.10(86) = 8.6	.10(92) = 9.2
Layout	.20	40	70	.20(40) = 8.0	.20(70) = 14.0
Operating costs	.15			.15(80) = 12.0	.15(90) = 13.5
	Total =100			70.6	82.7

Accept this location alternative

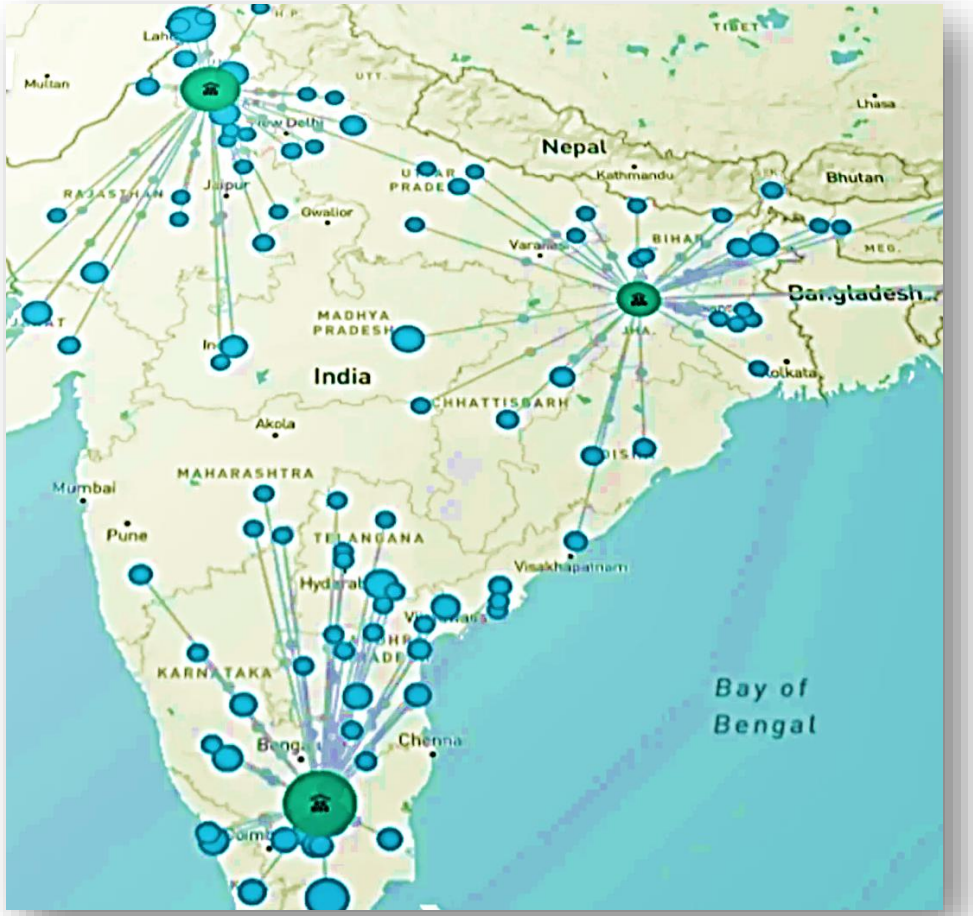
- Managers may choose to have a minimum threshold value (e.g., 85/100).
- If none of the locations meet the minimum, reject all.

Centre of Gravity Method

- The method use a precisely scaled map with co-ordinates of different locations or supply points [(x1,y1), (x2,y2), (x3,y3); etc] where supplies are delivered by the distribution center.

- The aim is to **minimize the travel time and/or transportation cost** of a distribution centre

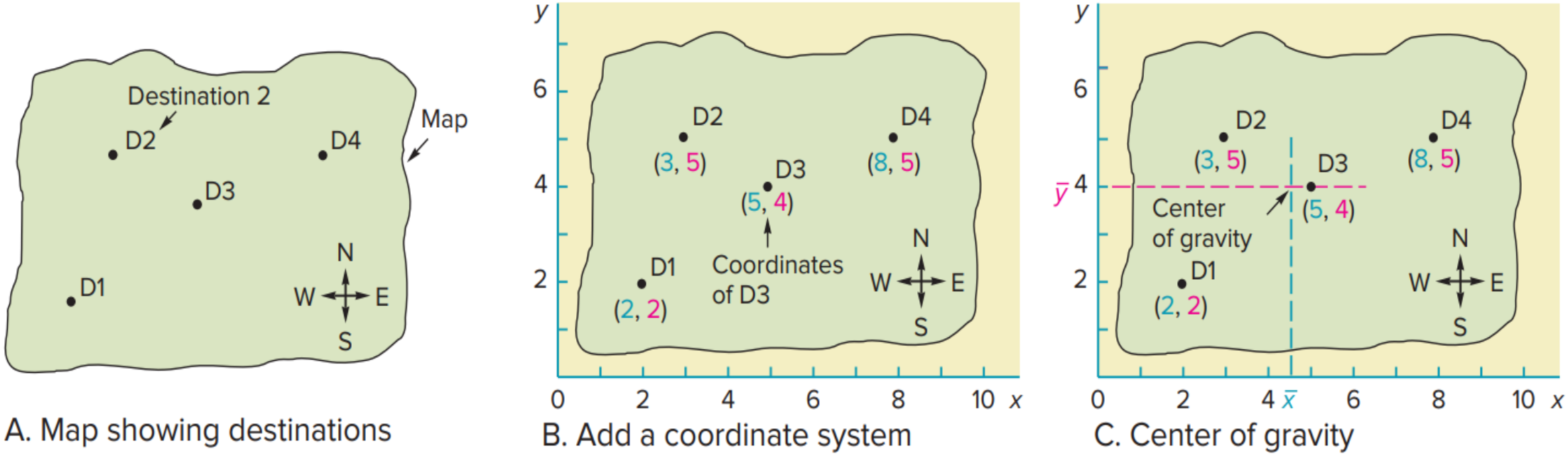
- Assumption:
  - Cost/unit distance is constant
  - Distribution cost is a linear function of the distance and the quantity shipped.



Centre of Gravity Method

Minimize the travel time and/or transportation cost of a distribution centre

Managerial Concern : Shall we take is a Minimization problem or Maximization Problem ?



A. Map showing destinations

B. Add a coordinate system

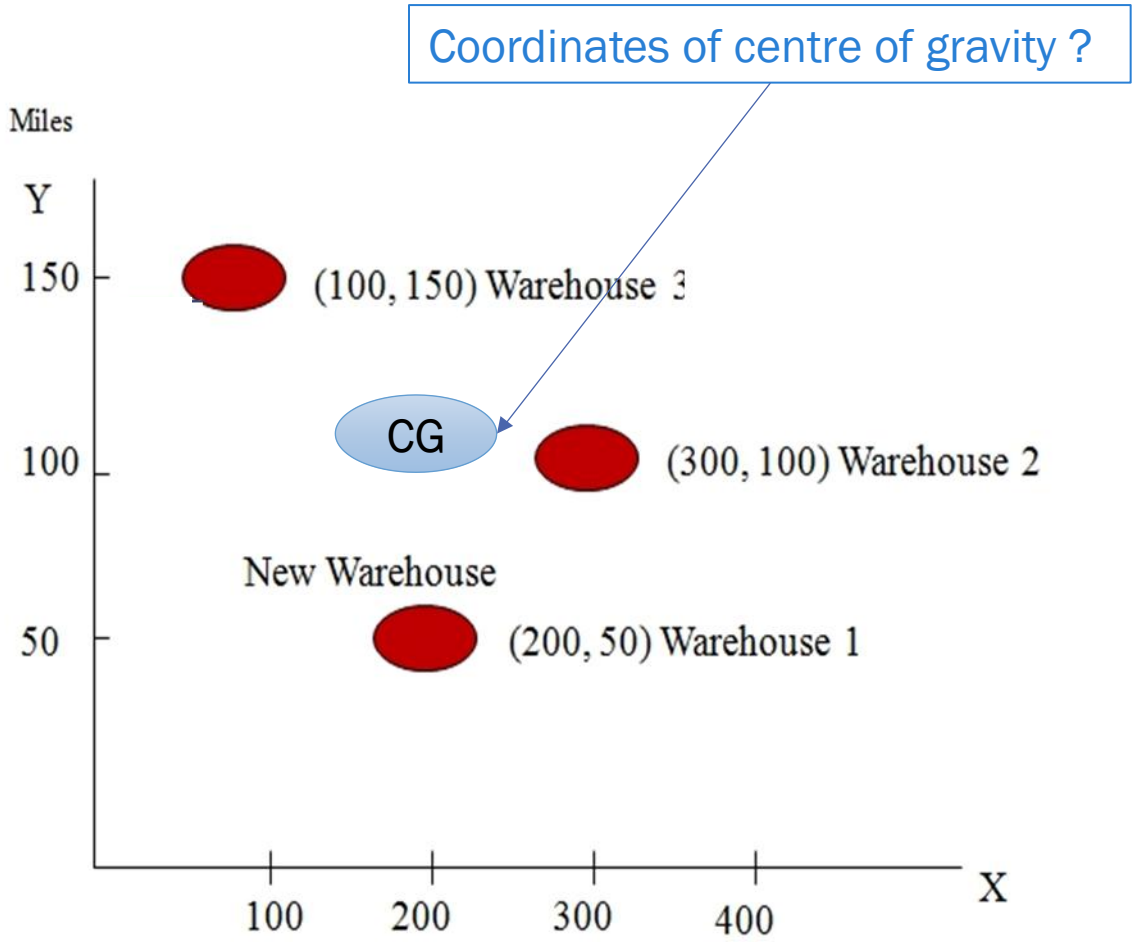
C. Center of gravity

Coordinates of centre of gravity ?

A company wants to expand its facility and locate a distribution centre within a network of three existing warehouses. Given the following assumptions below, what are the coordinates for the new potential distribution centre?

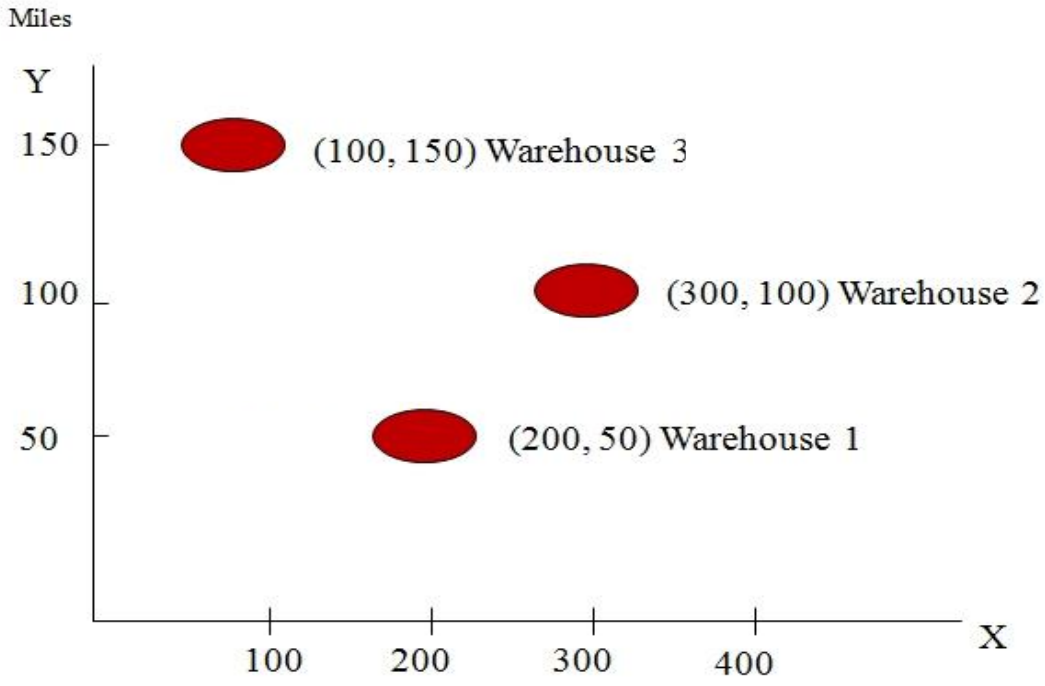
Given

- Warehouse 1 has a daily supply of 2,500 units
- Warehouse 2 has a daily supply of 1,300 units
- Warehouse 3 has a daily supply of 5,000 units



Centre of Gravity Method

Warehouse	X-coordinate	Y-coordinate	Quantity of Shipment
1	200	50	2500
2	300	100	1300
3	100	150	5000



Coordinates of Centre of gravity:

$$\bar{x} = \frac{\sum x_i Q_i}{\sum Q_i}$$

$$\bar{y} = \frac{\sum y_i Q_i}{\sum Q_i}$$

Cx= (200 x 2,500) + (300 x 1,300) + (100 x 5,000) / (2,500 + 1,300 + 5,000) = 158

Cy= (50 x 2,500) + (100 x 1,300) + (150 x 5,000) / (2,500 + 1,300 + 5,000) = 114

Coordinates of centre of gravity

$$\bar{x} = \frac{\sum x_i}{n}$$

$$\bar{y} = \frac{\sum y_i}{n}$$

when quantity of shipments (Q) are equal

where

$x_i = x$ coordinate of destination i
$y_i = y$ coordinate of destination i
$n =$ Number of destinations

$$\bar{x} = \frac{\sum x_i Q_i}{\sum Q_i}$$

$$\bar{y} = \frac{\sum y_i Q_i}{\sum Q_i}$$

when quantity of shipments (Q) are unequal

where

$Q_i =$ Quantity to be shipped to destination i
$x_i = x$ coordinate of destination i
$y_i = y$ coordinate of destination i

Locational CVP (Cost-Volume-Profit) Analysis

- CVP analysis helps in determining the **break-even point (BEP);** **a point beyond which a business investment starts making profit.**
- It analyse how changes in **production quantity** (Q), **costs (C), and prices (P)** to stay profitable.
- It helps in comparing the locational alternatives.

How to conduct CVP analysis:

- Determine the key **cost variables** associated with each location alternative.
- Determine which location will have the lowest total cost (or highest profit) for the expected level of output
- Further analysis.

CVP Analysis

- Key variables:
  - *F* : *Fixed costs (for the period)*
  - *v* : *Variable cost per unit*
  - *P* : *Selling price per unit*
  - *Q* : *Quantity (units sold)*

- **Fixed Costs (FC)** –

  Land lease cost (periodic), Construction/renovation cost, Property tax. Licensing fee, Insurance, etc.

- **Variable Costs(v):**

  Labour cost, Raw Material Cost, Import Duties/Taxes, Water & Electricity Cost, Transportation cost, etc.

Variable Cost components depend on the produced quantity

CVP Analysis

- *F* : *Fixed costs (for the period)*
- v : Variable cost per unit
- P : Selling price per unit
- Q : Quantity (units sold)

Profit = Total Sales Revenue – Total Cost

= Total Sales Revenue – (Fixed Cost + Variable Cost)

= P*Q – (F + v*Q)

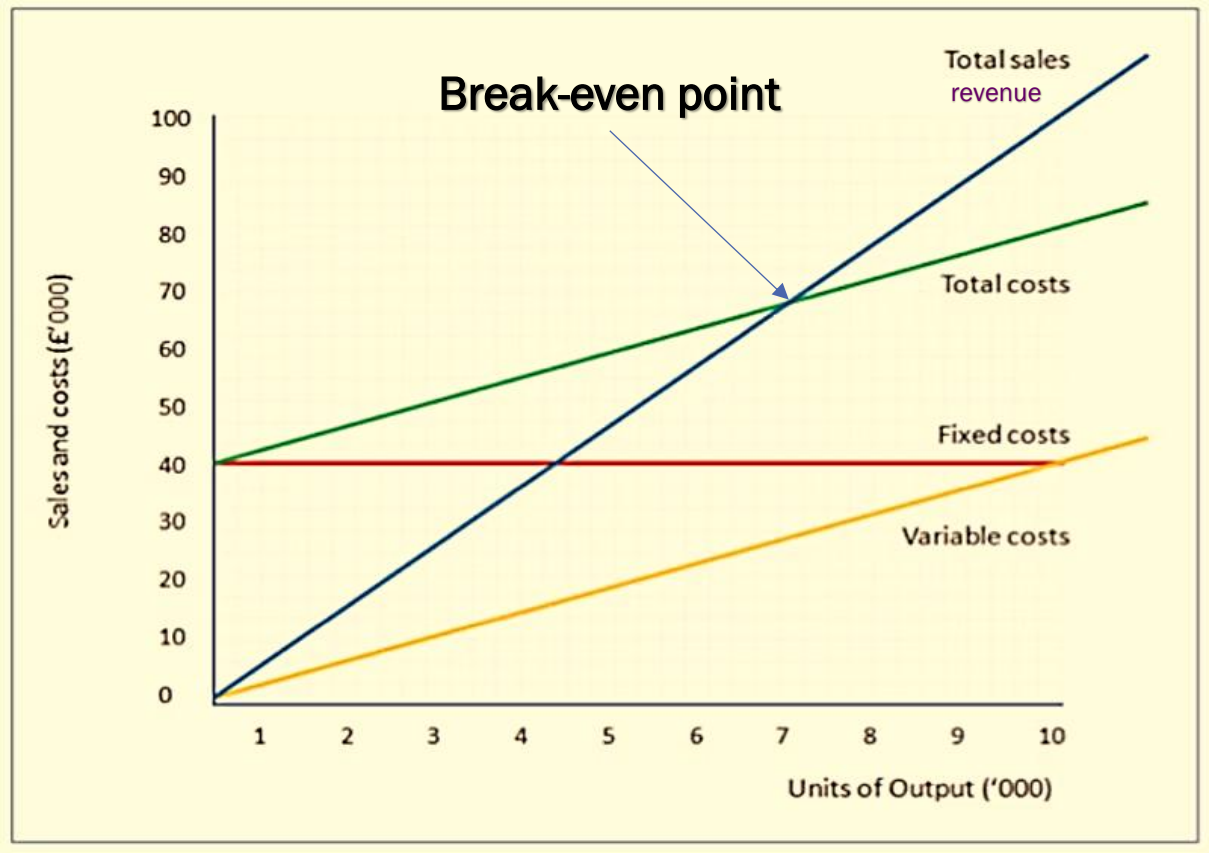
= (P−v)Q−F

= $CM_u \cdot Q - F$

- CM_u = Contribution margin/unit= P – v

CVP Analysis: Break Even Point

- Total Costs = Fixed Cost + Variable Cost
  = *F* + *v*Q*
- Total Sales Revenue = P*Q
- Break Even Point:
  - *Total Costs = Total revenue*
  - *i.e., Profits = 0*



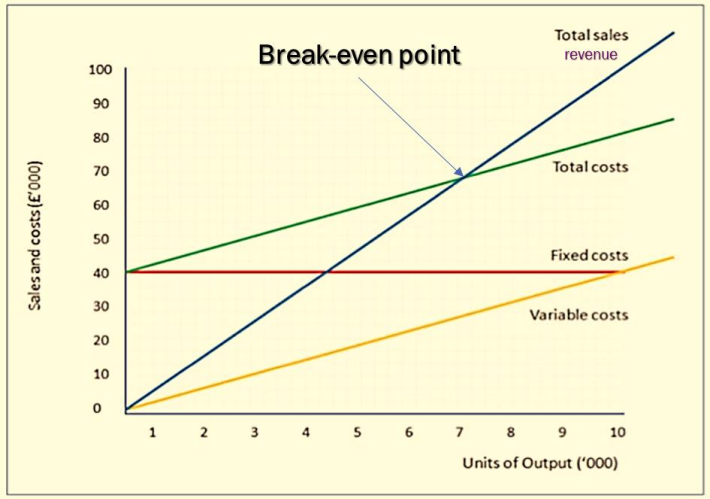
CVP Analysis: Break Even Point

- Break Even Point:

$$\textit{Profits} = (P-v)Q-F = 0$$

$$\Rightarrow \quad Q_{BE} = F / (P-v) = F / CM_u$$

- CM_u = Contribution margin/unit= P – v



Example: CVP analysis

- A business unit has to expand its operations and it is considering two different locations, one in urban and the other in suburban region for establishing its new production facility. The cost figures at these locations are as follows:

Location	Fixed Cost (FC)	Variable Cost (VC)
Urban	$400,000/yr	$22/unit
Suburban	$250,000/yr	$28/unit

Selling price (P) per unit: $50 (same for both locations)

Contribution margin (CM) = P – v

- Urban CM = 50 – 22 = **$28/unit**
- Suburban CM = 50 – 28 = **$22/unit**

Example contd…

Location	Fixed Cost (FC)	Variable Cost (VC)
Urban	$400,000/yr	$22/unit
Suburban	$250,000/yr	$28/unit

Selling price (P) per unit: $50 (same for both locations)

Contribution margin (CM) = P − v

- Urban CM = 50 − 22 = **$28/unit**
- Suburban CM = 50 − 28 = **$22/unit**

Breakeven Q_{BE} = F ÷ CM

- **Break Even for Urban Location** : 400,000 ÷ 28 = **14,286 units**
- **Break Even for Suburban Location**: 250,000 ÷ 22 = **11,364 units**

Comment:

Suburban breaks even sooner (lower fixed cost), Urban needs more volume to cover its higher fixed cost.

Decision: ?

Example:

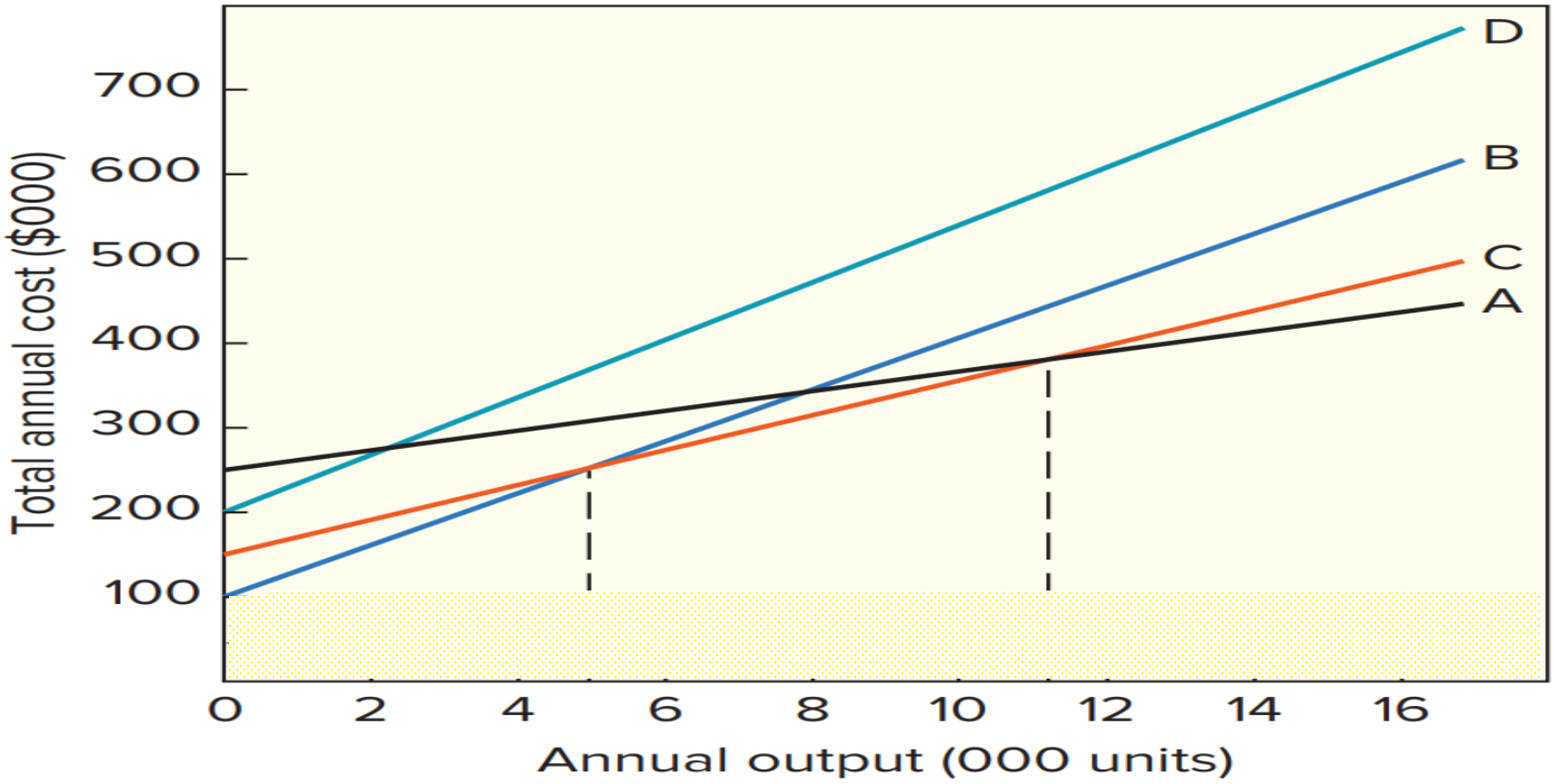
Fixed and variable costs for four potential plant locations are shown as follows.

Location	Fixed Cost per Year	Variable Cost per Unit
A	$250,000	$11
B	100,000	30
C	150,000	20
D	200,000	35

a. Plot the total-cost lines for these locations on a single graph.

b. Identify the range of output for which each alternative is superior (i.e., has the lowest total cost).

c. If expected output at the selected location is to be 10000 units per year, which location would provide the lowest total cost?


D
B
C
A
700
600
500
400
300
200
100
Total annual cost ($000)
0
2
4
6
8
10
12
14
16
Annual output (000 units)

Which location is more profitable for a unit sales of 10000/year ?

	Fixed Cost	+	Variable Cost	=	Total Cost
A	$250,000	+	$11(10,000)	=	$360,000
B	100,000	+	30(10,000)	=	400,000
C	150,000	+	20(10,000)	=	350,000
D	200,000	+	35(10,000)	=	550,000

Evaluating Location Alternatives: **Quantitative Approaches**

- Factor Rating (also incorporates subjective criteria)
  - *Focuses on production cost efficiency.*
  - ***Best for:*** *Multi-criteria decision-making when both qualitative and quantitative factors are important.*
- The Centre of Gravity Method
  - *Optimizes Logistics & distribution*
  - ***Best for:*** *Warehouses, distribution centres, and logistics-heavy businesses.*
- Cost-Profit-Volume Analysis (Break-Even Analysis)
  - ***Best for:*** *Companies wanting to assess the cost-effectiveness of locations at different production volume*